

Milestone 3

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StepQuest

Official Logo and App Icon



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Main App Idea:

StepQuest is an app inspired by Duolingo, it was built around the idea of making walking a fun and rewarding activity, by implementing gamification.

The app tracks the user's steps by using Step Counter Sensor and Validates the data by using three additional sensors to prevent cheating and abusing the step counter sensor. The user steps are used to unlock badges, complete challenges and participate in a ranking system which is reset every week and rewards the user with experience and coins. The user can view other user that are competing in the same rank, which create a competitiveness between user to aim for high step counts.

The user can also maintain a streak by reaching their daily goal steps. The daily step goal increases progressively as the user levels up.

Milestone 3 Goals and objectives:

In this third Milestone, my main objective was to have a fully functional app which includes:

- Implement real database to save and load user data and statistics
- Building the app logic behind level, experience and rank progression
- Implement the streak logic with option to repair the streak.
- Implement the rank logic with rank promotion and weekly rewards for the top three
- Implement the badge logic to give the user badges after completing the required challenges
- Updating the user profile to reflect the user's stats
- Implement privacy policy page
- Implement terms and conditions page
- Adding 3 more sensors that are used to validate the user steps
- Optimizing the app and following best practices
- Running real user tests and collecting feedbacks

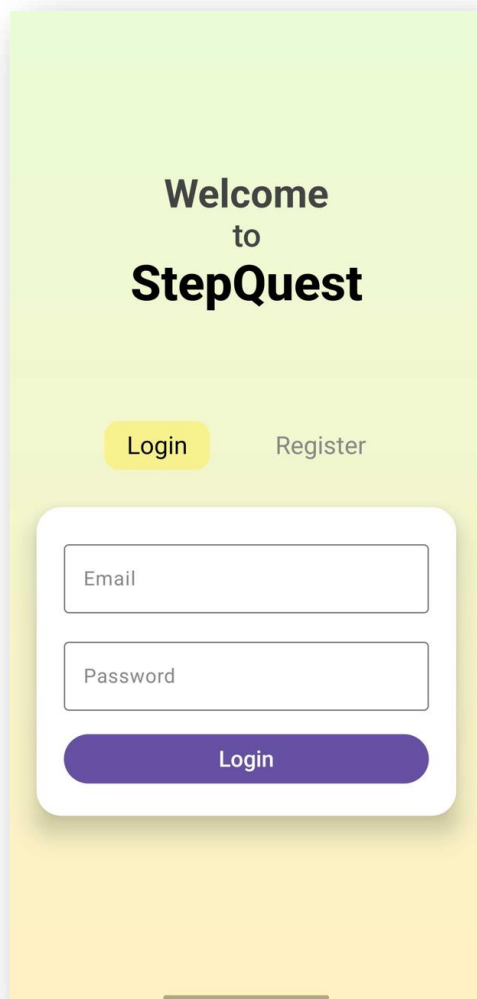
As well as making minor changes in the app design.

App Layout:

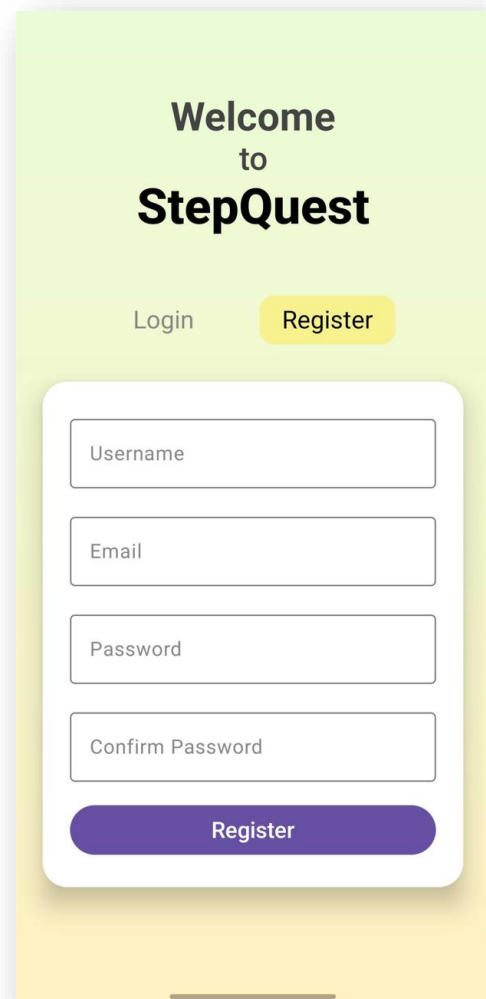
Auth Screen:

This is the first screen the user sees when they open the app for the first time, the user has to options Login or Register, the app uses Fire Base authentication and error detection, the app includes other validations like missing fields...

After the user logs in or registers, their data is saved in the database and they are redirected to the Home screen



Mockup of the Login screen. The background is a light green gradient at the top, transitioning to a light yellow gradient at the bottom. The text "Welcome to StepQuest" is centered in bold black font. Below the text, there are two buttons: "Login" (yellow) and "Register" (grey). Below these buttons is a white rounded rectangle containing two input fields: "Email" and "Password". At the bottom of this white rectangle is a purple button labeled "Login".



Mockup of the Register screen. The background is a light green gradient at the top, transitioning to a light yellow gradient at the bottom. The text "Welcome to StepQuest" is centered in bold black font. Below the text, there are two buttons: "Login" (grey) and "Register" (yellow). Below these buttons is a white rounded rectangle containing four input fields: "Username", "Email", "Password", and "Confirm Password". At the bottom of this white rectangle is a purple button labeled "Register".

Home Screen:

The Home Screen was built around simplicity while providing the user with the necessary information,

The user can easily:

- Track their daily steps count
- view their streak
- view the amount of coins owned
- Track their weekly, monthly total steps
- Track their daily step count for the current week

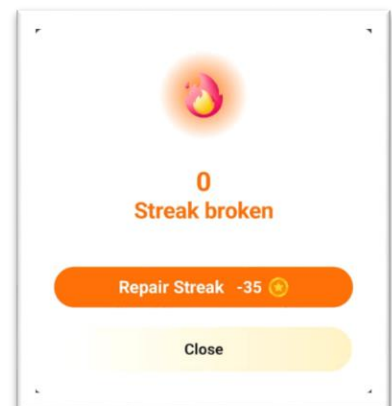
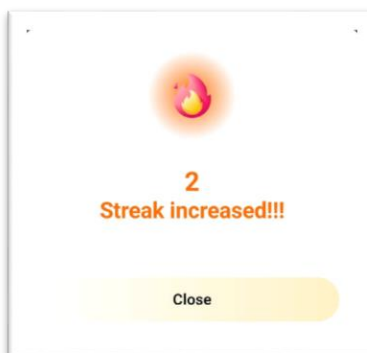
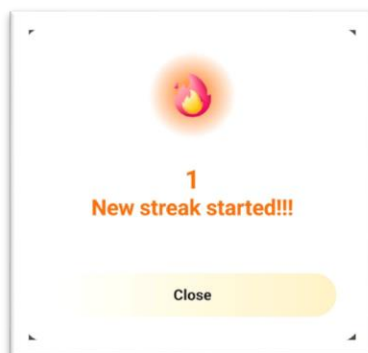
From this Activity the user can navigate to multiple other Activity using the nav bar, or by clicking on their profile picture.

The user can receive different popups in this Activity:

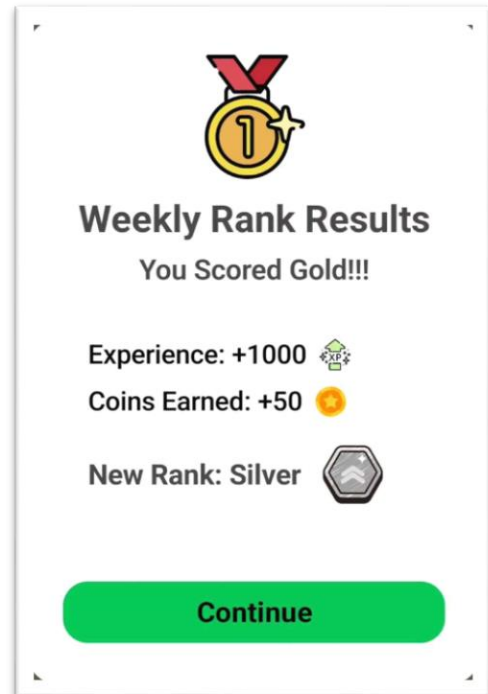
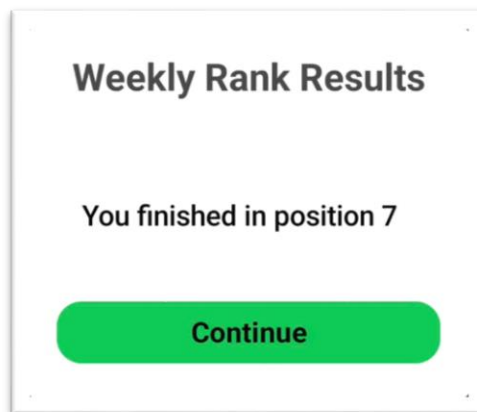
Streak Related Popups notifying the user of either

- Streak Increased
- New Streak Started
- Streak Broken

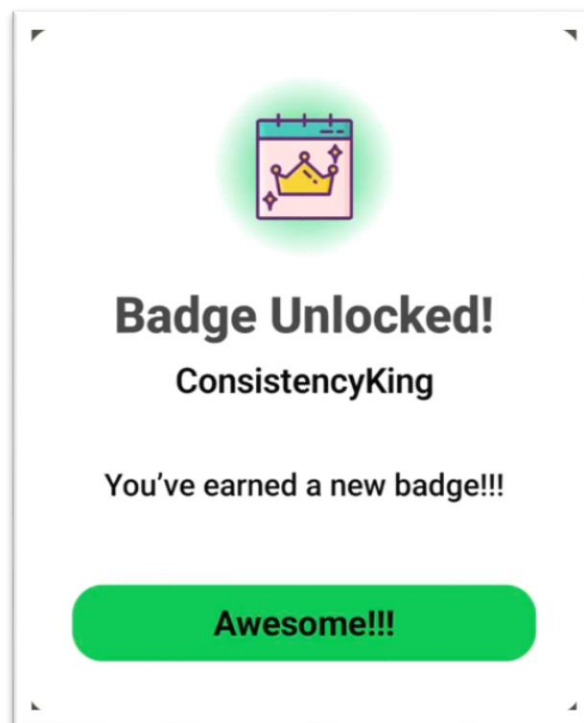
Streak broken popup gives the user the option to repair the streak by consuming coins



Other Popups include Weekly rank Results Which notify the user of the position they achieved and the rewards they obtained if any.



The user also gets notified when they obtain a new badge, after completing the required challenge.



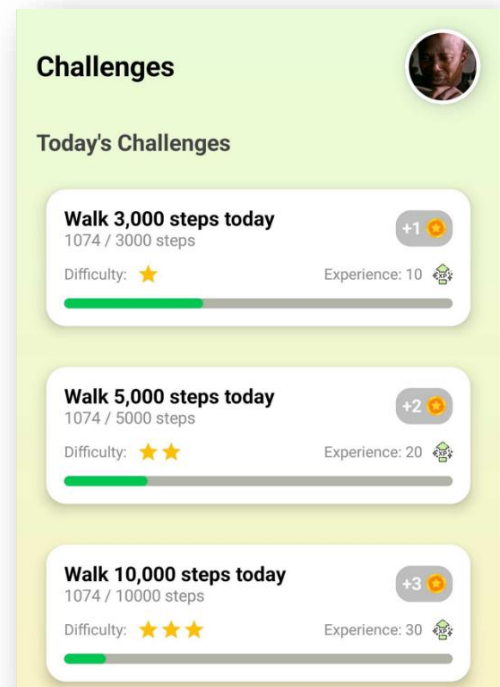
Challenges Screen:

In this activity the user can view their Daily challenges, and some tips.

The challenges cards were designed in a way to allow the user to easily view the challenge requirement, the difficulty, and rewards including coins and experience.

The user can also easily track their progress in each challenge with the progress bar.

After a challenge is completed the Reward Icon becomes green which indirectly tells the user that the reward is now claimable, once the user claims the reward the experience and coins gets added to their profile in the database



Badges Screen:

in this activity the users can view the badges they have obtained and the badges that are still locked.

The user can click on each badge to view the requirements to obtain that badge,

Badges' challenges could be time box as an example Nightwalker would require the user to make 2000 steps after 9pm, other badges might be related to the user's streak like the consistency king which is given to a user once they have accumulated a 7 days streak.

My initial plan was to make a user's badges viewable in their profile as well so other user can see them, but unfortunately time was not in my favor and making profiles public was not achieved this milestone. But I will be working on it on my free time to make it possible.



Rank Screen:

This Activity allows the user to view the current rank they are competing in, as well as viewing the current leader board and where they stand.

The ranking system gets reset every week on Monday, last week's results are calculated and users receive rewards depending on their rank.

only the first three ranks are rewarded Coins and experience, other positions do not receive any Rewards.

The holder of the first position in the leader board gets promoted to the next rank, and gets to compete against more competitive users.

The users receive medals as well depending on their position In the top 3, these medals can be viewed in the profile page

Profile Screen:

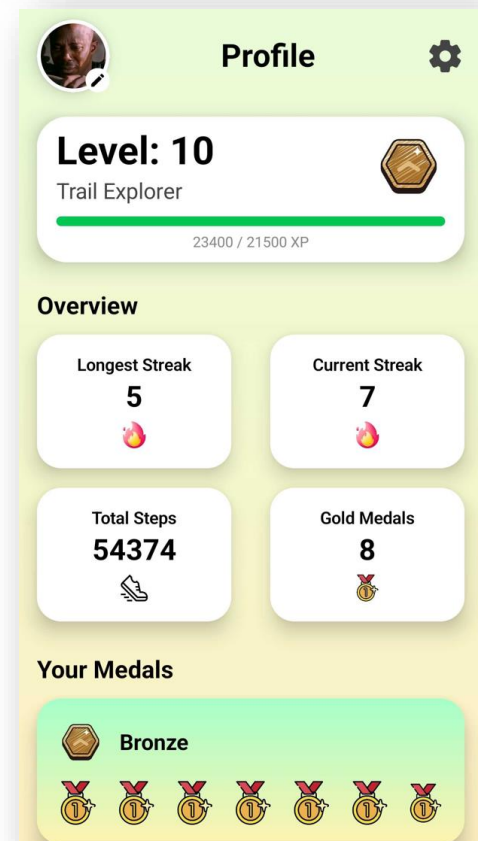
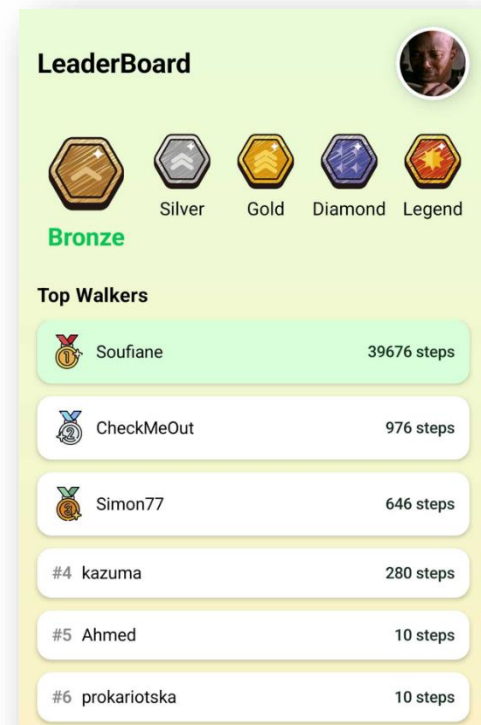
In this screen the user can change their profile picture by clicking on their current profile picture, the picture they used is stored locally (I was not able to get fire base storage).

The user can view their current rank, level, title and the required experience to reach the next level.

An Overview section allows the user to view their longest Streak, current streak, total steps and the number of gold medals acquired.

the user can view the different medals they have acquired in every rank

From this Activity the user can navigate the settings.



Settings Screen:

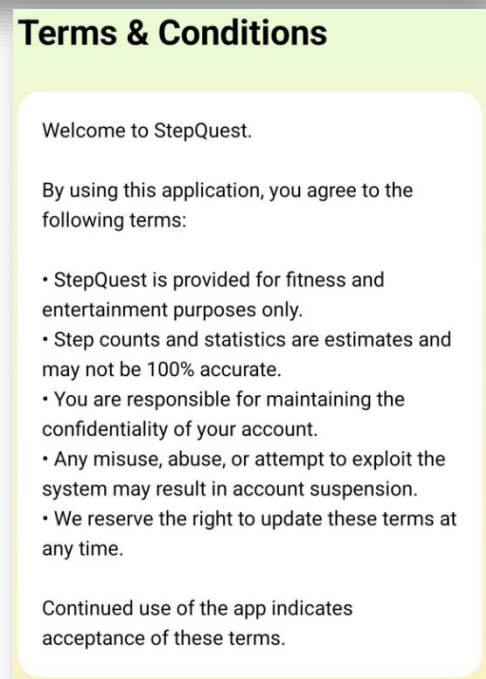
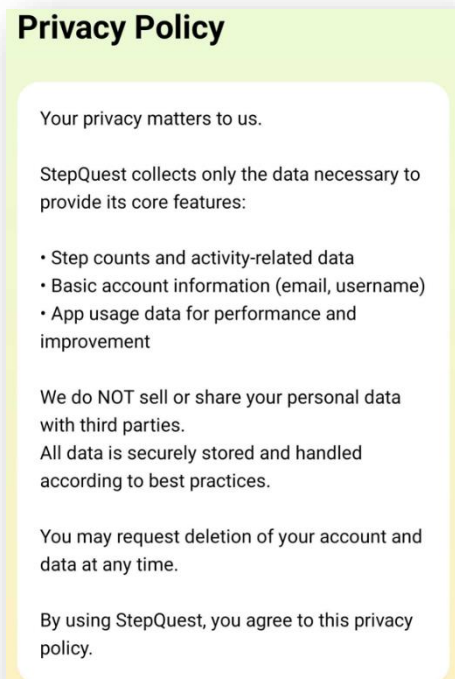
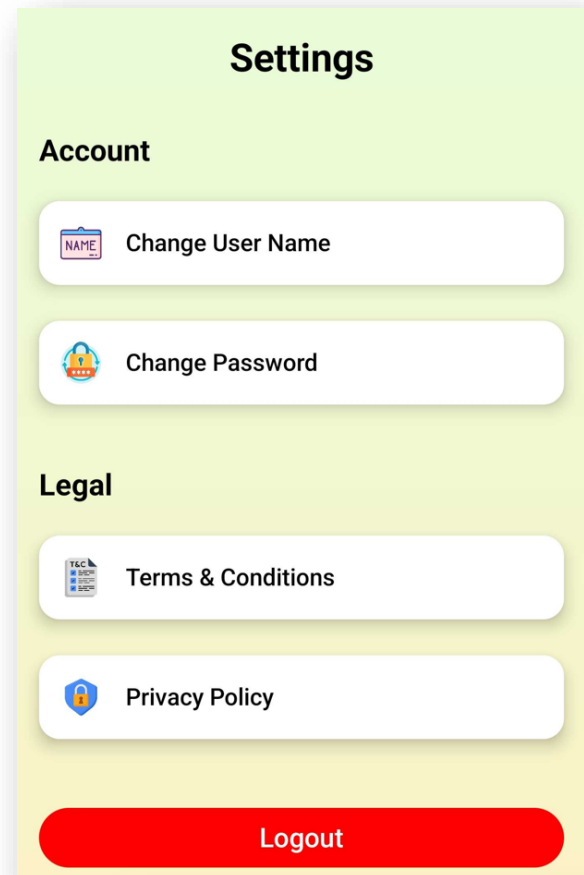
This screen allows the user to change their user name, and their password. These functionalities have been demonstrated in the previous milestone the only changes done is the implementation of the database.

The user can logout from their account from this activity.

Terms and conditions as well as privacy policy pages have been implemented.

Terms and condition content was influenced from **(FIP, 2025)**

Privacy and policy was influenced from **(fitnessonline, 2025)**



App Sensors:

The app currently uses 4 sensors:

TYPE_STEP_COUNTER

As explained by **(Developer.android, 2025.)** is a sensor that counts the user's steps, StepQuest calculates the daily steps by applying an offset to the raw data provided by the sensor, I have noticed that this sensor is very naïve and can easily be abused by shaking the phone or moving it in a certain way, which can allow the user to accumulate hundreds of steps in matter of seconds, I have found that other applications like Samsung health allow these type of behavior but that is because the app is not meant to be a game nor a competition.

For StepQuest this type of behavior can not exists and so more sensors were implemented to validate the data

TYPE_ACCELEROMETER

As explained by **(Hanly, 2025.)** This sensor measures the total acceleration including the gravity and is used to detect unusually strong or sudden movements. The app calculates the acceleration magnitude and rejects steps when the value exceeds a threshold.

High spikes usually indicate shaking, dropping the phone, or aggressive movement rather than normal walking.

TYPE_LINEAR_ACCELERATION

As explained by **(Developer.android, 2025.)** This sensor provides acceleration data without gravity, which helps to detect linear motion. The app calculates the magnitude of this signal and blocks steps that have sharp linear acceleration spikes

TYPE_GYROSCOPE

As explained by **(Developer.android, 2025.)** The gyroscope measures rotational motion, it was used in StepQuest to detect hand swinging, shaking or tilting the phone. While observing recent stored rotation values, a rotational pattern was noticed when trying to generate fake steps using those movement, when this pattern appears repeatedly within a short time window the steps generated are ignored.

Optimization

Step handling logic was separated into its own StepCounter class . This class is responsible only for sensor management, step calculation and validation which keeps this logic out of the activities.

To optimize the app performance and battery usage, all sensors are now processed on a separate background thread instead of the main UI thread. The sensors are registered using normal polling rates. The step counter sensor remains the primary source of steps due to its low power usage, while the other sensors are only used to validate data rather than calculating steps. Sensors are also registered and unregistered correctly based on the activity lifecycle.

Data loading is also done in a specific order to ensure everything is loaded correctly.

Most of the logic and data handling is moved into ViewModels to keep the UI code clean and lightweight, as composables only that reacts to state changes instead of handling logic directly. Entire screens state values were stored using mutableStateOf inside ViewModels to allow Compose to update only the parts of the UI that actually change, instead of redrawing the entire screen and reloading the data when navigating between screens.

Icons used across the app were separated into an IconsMapping utility. Instead of hardcoding icon resources in multiple places, ranks, medals and badges were mapped to their icons in one location.

Data Base:

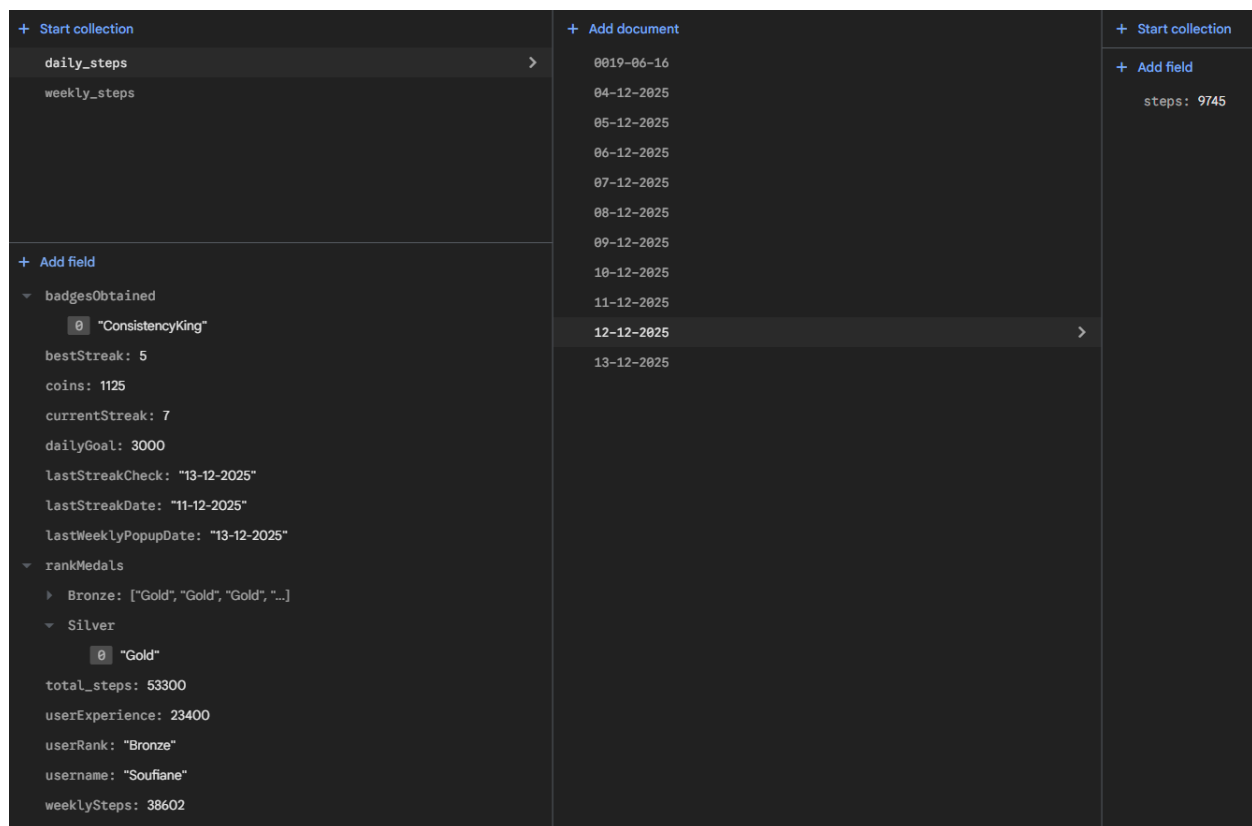
For the database Firebase was used to store the users data and handle user authentication.

Firebase was chosen mainly because it is simple to use and easy to integrate, which allowed the app to be developed without having to create or manage a custom backend or server.

Firebase also provides built-in tools for testing, monitoring and debugging, which made it very useful during development and testing stages.

The platform was also used for app distribution and to receive feedback from users while testing StepQuest.

Example of data stored:



The screen shot only includes my own data

Bibliography

FIP, M.K.C., CIPM, CIPT. (2025) *Mobile App Terms and Conditions Template*. Termly. Available at: <https://termly.io/resources/templates/app-terms-and-conditions/> (Accessed: 13 December 2025).

Hanly, S. *Accelerometers: Taking the Guesswork out of Accelerometer Selection*. Available at: <https://blog.endaq.com/accelerometer-selection> (Accessed: 13 December 2025).

Linear Acceleration - an Overview | ScienceDirect Topics. Available at: <https://www.sciencedirect.com/topics/computer-science/linear-acceleration> (Accessed: 13 December 2025a).

Motion Sensors | Sensors and Location. Android Developers. Available at: https://developer.android.com/develop/sensors-and-location/sensors/sensors_motion (Accessed: 13 December 2025b).

Privacy Policy. Available at: <https://fitnessonline.app/privacypolicy.html> (Accessed: 13 December 2025c).

Ready to Start Tracking Your Steps? You Can Count on These Editor-Tested Pedometers. Verywell Fit. Available at: <https://www.verywellfit.com/best-pedometers-4159148> (Accessed: 13 December 2025d).

What Is Gyroscope in Phones? (How to Use) - HONOR AE. Available at: <https://www.honor.com/ae-en/blog/what-is-gyroscope-in-phones/> (Accessed: 13 December 2025e).